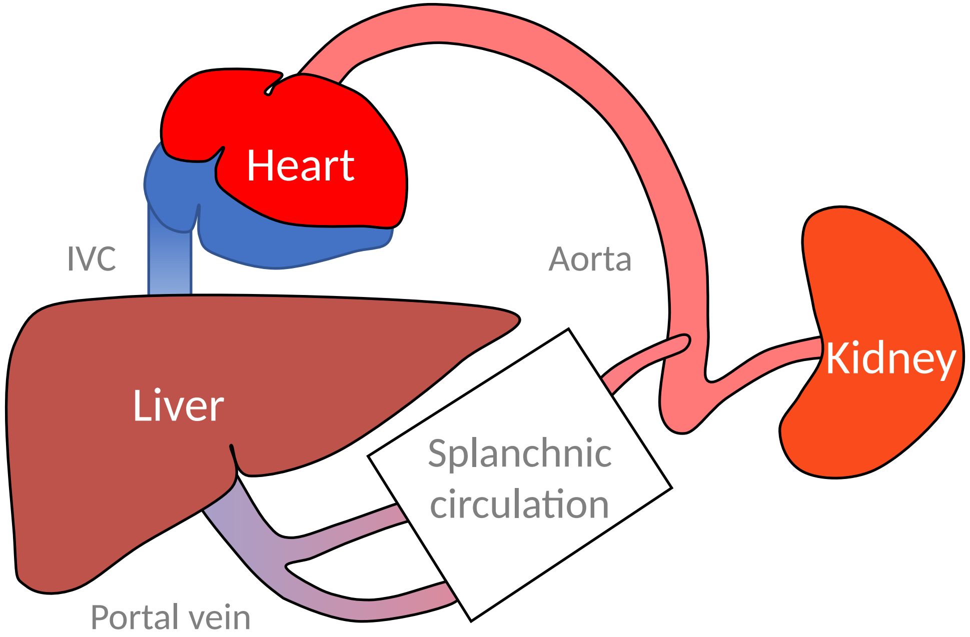
Differential Diagnosis of AKI in ESLD

Differential
Pathophys
Dx/Tx

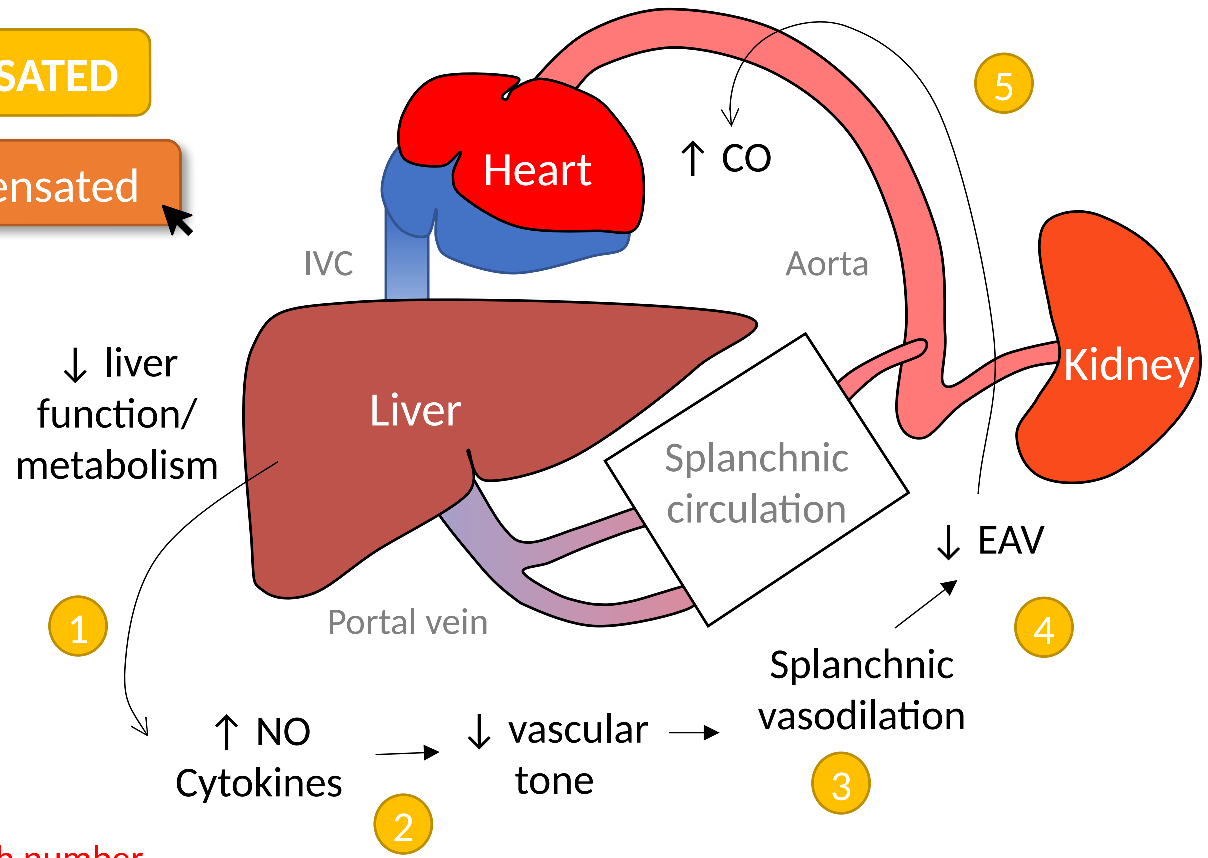


60-70%	PRERENAL	↓ volume (GI bleed, overdiuresis, GI losses) Sepsis (SBP, pneumonia, UTI) Medications	*Volume responsive*
		HRS Cardiorenal	*Not volume responsive*
30-40%	INTRARENAL	ATN AIN Glomerulonephritis	
<1%	POSTRENAL	Obstruction (intrinsic, extrinsic) Neurogenic bladder	

Pathophysiology of Hepatorenal Syndrome

Differential
Pathophys
Dx/Tx

COMPENSATED
Decompensated



Click each number

In compensated cirrhosis, ↓ liver function leads to:

1. ↓ metabolism of vasodilatory substances which results in
2. ↓ vascular tone throughout the body including the splanchnic circulation (which supplies the gut)
3. Splanchnic vasodilation results in pooling of blood in the splanchnic circulation
4. This ↓ effective arterial volume (EAV)
5. The heart increases cardiac output in response to ↓ EAV which preserves blood pressure and organ perfusion

Pathophysiology of Hepatorenal Syndrome

Differential
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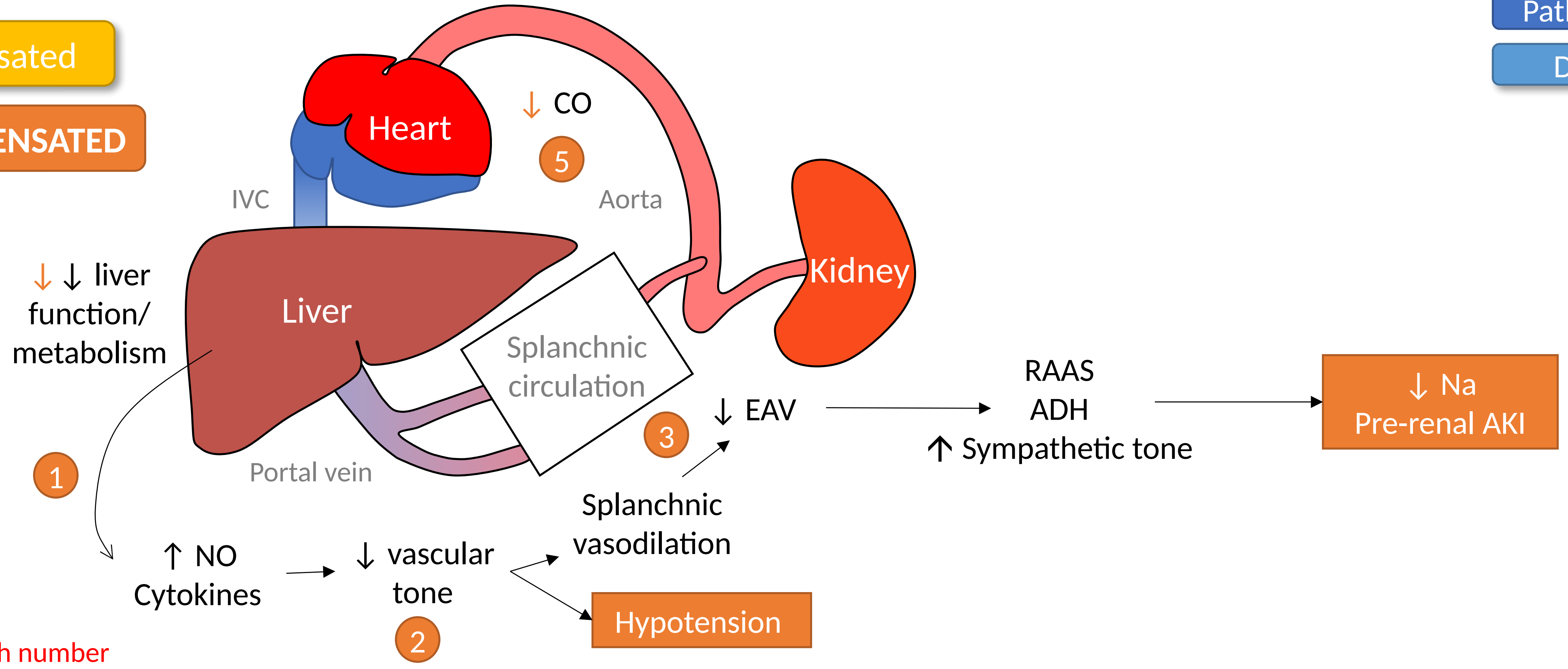
Compensated

DECOMPENSATED



Click each number

In DEcompensated cirrhosis, ↓ liver function leads to:

1. ↓ metabolism of vasodilatory substances which results in
2. ↓ vascular tone throughout the body including the splanchnic circulation
3. Splanchnic vasodilation results in pooling of blood in the splanchnic circulation, resulting in ↓ EAV
4. This ↓ EAV activates RAAS, ADH, and ↑ sympathetic tone leading to hyponatremia and AKI
5. Cardiac output fails to compensate leading to a ↓ in BP leading to pre-renal AKI and hypotension

Diagnosis of Hepatorenal Syndrome

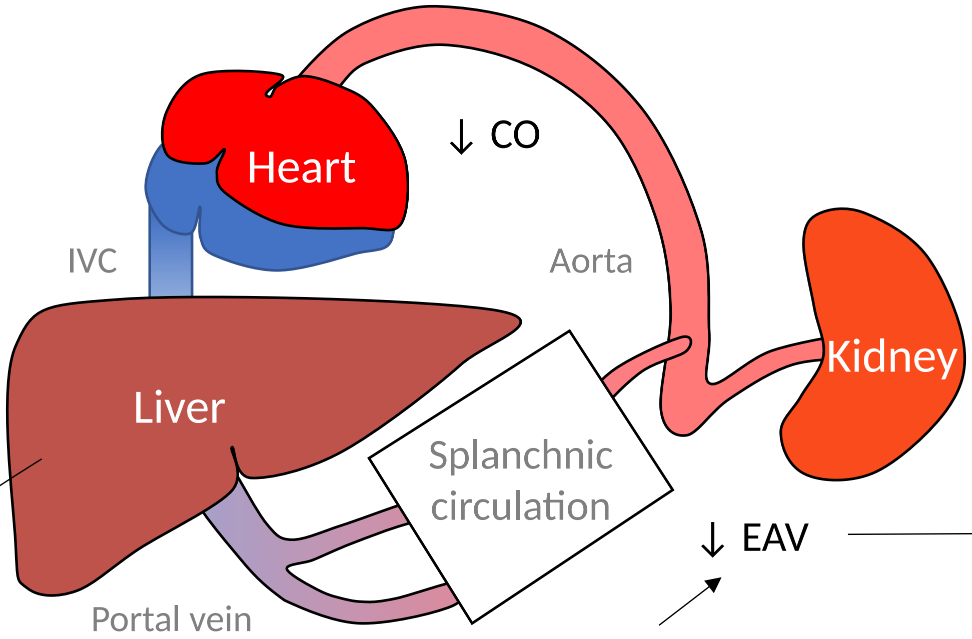
Differential

Pathophys

Dx/Tx

<u>DE</u>Compensated cirrhosis

↓↓ liver function/ metabolism



↑ NO
Cytokines → ↓ vascular tone → Splanchnic vasodilation → ↓ EAV → RAAS / ADH / ↑ Sympathetic tone → ↓ Na / Pre-renal AKI

↓ vascular tone → Hypotension

TREATMENT

DIAGNOSIS

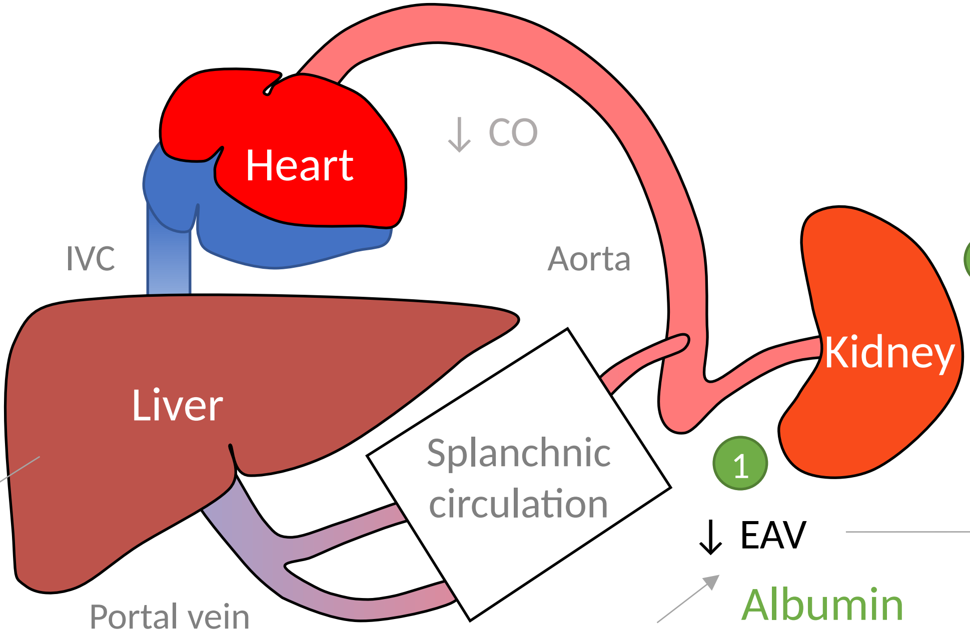
Pre-renal AKI → Albumin challenge (1 g/kg x 48 h) STOP diuretics → No Δ in Cr = HRS

		ONSET	PROGNOSIS
Type 1	↑ Cr > 2.5 or ↓ GFR < 20 mL/hr	<2 weeks	~ 2 wk survival
Type 2	Slow ↑ in Cr to 1.5 – 2.5	<2 weeks	~4-6 mos survival

Treatment of Hepatorenal Syndrome

Differential
Pathophys
Dx/Tx

DECompensated cirrhosis



5 Dialysis

TIPS Transplant 4 ↓↓ liver function/ metabolism

1 ↓ EAV → RAAS ADH ↑ Sympathetic tone → ↓ Na Pre-renal AKI

Albumin

↑ NO Cytokines → ↓ vascular tone 2 → 3 Splanchnic vasodilation

Octreotide

Hypotension

Vasopressors

Click each number

TREATMENT

1. Albumin
2. Vasopressors (norepinephrine, midodrine) to ↑ MAP > 15
3. Octreotide – ↓ splanchnic dilation
4. TIPS
5. HD as bridge to liver transplant, which is ultimate treatment

DIAGNOSIS

Pre-renal AKI → Albumin challenge (1 g/kg x 48 h) STOP diuretics → No Δ in Cr = HRS

HRS → Type 1: ↑ Cr > 2.5 or ↓ GFR < 20 mL/hr

HRS → Type 2: Slow ↑ in Cr to 1.5 – 2.5

Take-Home Points

1

AKI in cirrhosis is usually pre-renal

Pre-renal causes account for 60–70%; obstruction is <1%. Always work up infection (diagnostic paracentesis for SBP) and rule out GI bleed.

2

HRS = pre-renal AKI that doesn't respond to volume

Diagnose by no improvement in Cr after stopping diuretics + 48-h albumin challenge (1 g/kg/day), with no shock, nephrotoxins, or structural kidney disease.

3

Albumin + a vasoconstrictor is the medical core

Albumin to raise effective arterial volume + a vasoconstrictor to restore tone. Definitive treatment is liver transplant; HD/TIPS are bridges in selected patients.

4

Treat SBP with albumin too

In SBP, albumin 1.5 g/kg day 1 + 1 g/kg day 3 reduces HRS and mortality (esp. Tbili >4, Cr >1, BUN >30).

About This Optimized Edition

Original: “AKI in Cirrhosis” — TeachIM.org (Yilin Zhang, MD; Univ. of Washington / Univ. of Colorado). Published ~2020–2021.

Correction in this edition: Treatment slide “midorine” → “midodrine”. All animations and graphics preserved.

Guideline update — terlipressin: The original teaching script states terlipressin is “available in other countries but not in the US.” As of Sept 2022 (CONFIRM trial) the FDA has approved terlipressin for HRS-AKI — the first drug approved for HRS in the US. Avoid in ACLF grade 3, hypoxia, or SCr >5 mg/dL (respiratory-failure signal).

Guideline update — nomenclature: “Type 1 / Type 2 HRS” has been replaced (ICA 2015; AASLD 2021): HRS-1 → HRS-AKI, diagnosed by ICA-AKI criteria (↑Cr ≥0.3 mg/dL in 48 h or ≥50% in 7 d) rather than the old fixed Cr >2.5 threshold; HRS-2 → HRS-NAKI. The deck’s schematic is retained for teaching; use the modern terms when you present.

Companion files: Lesson package (HTML), one-pager (PDF), Anki cards, learner board, case handout, and case answer sheet — all in this lesson folder.